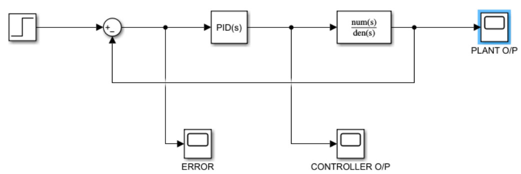
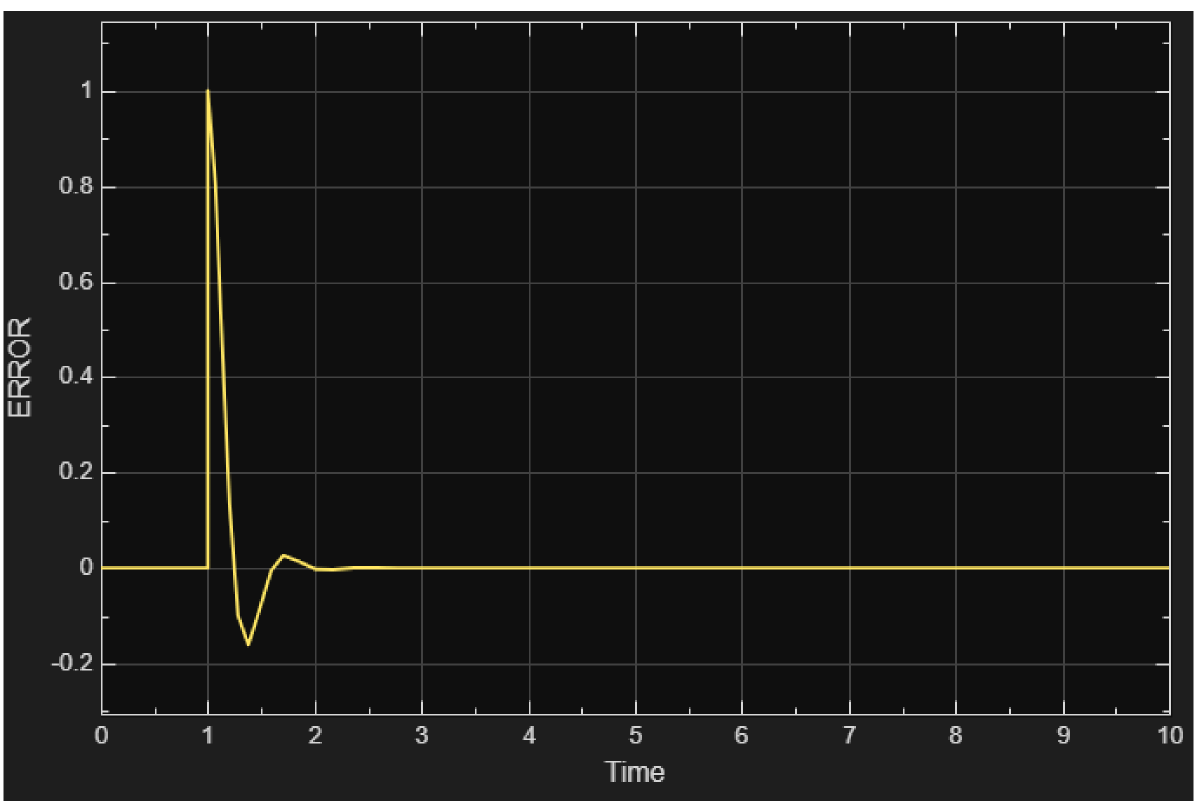
DC motor with PID control (transfer function =s^2 +2s+1)



P=1

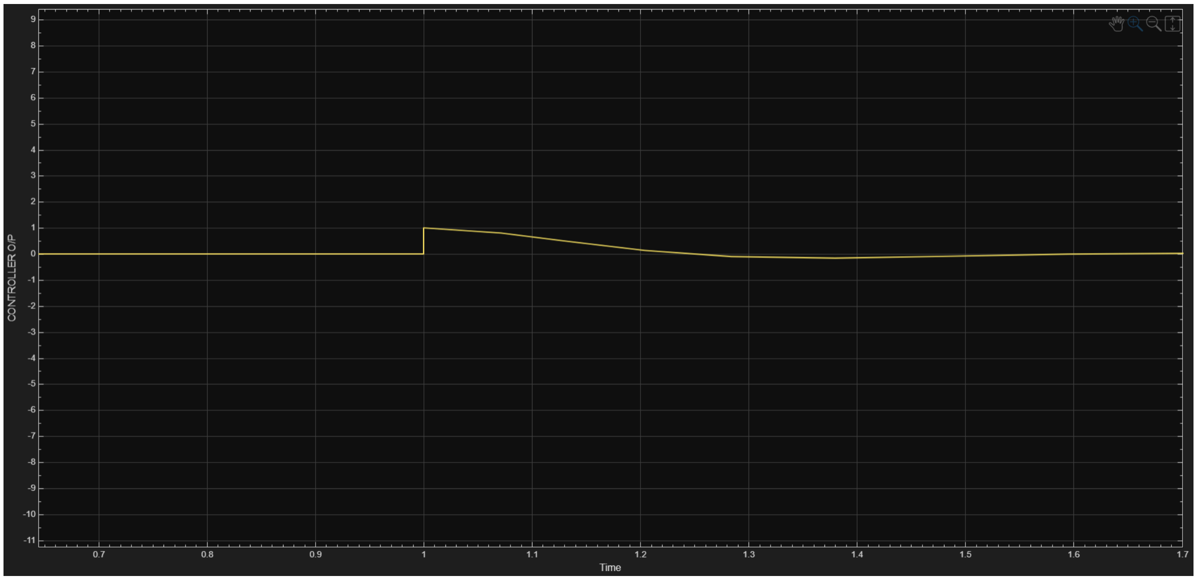
Screenshots

1. ERROR

    - 

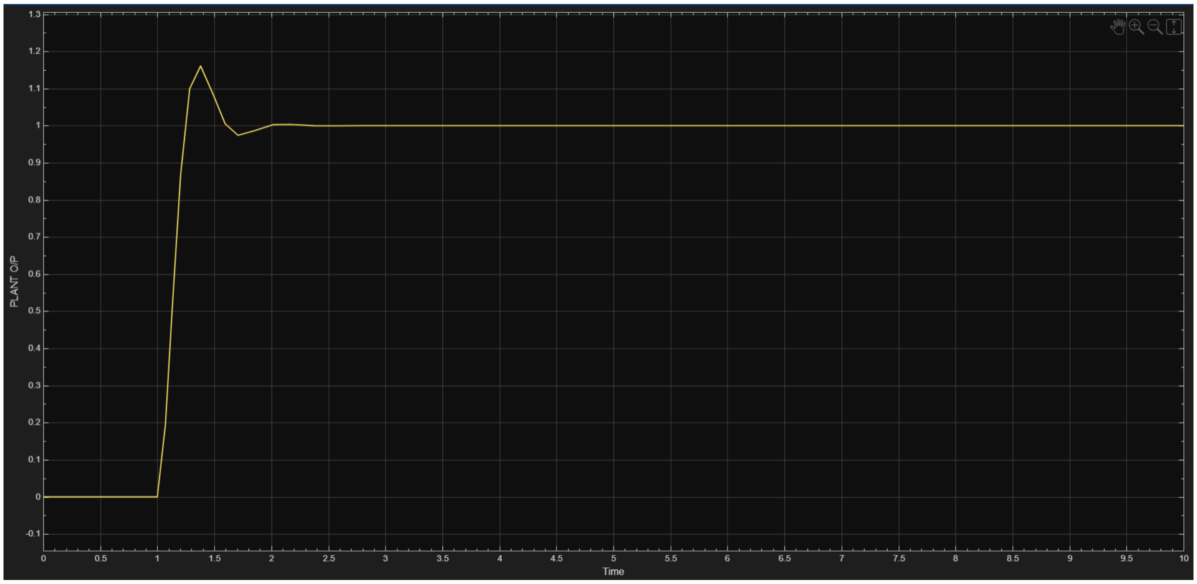
2. CONTROLLER O/P

-



3. PLANT O/P

-

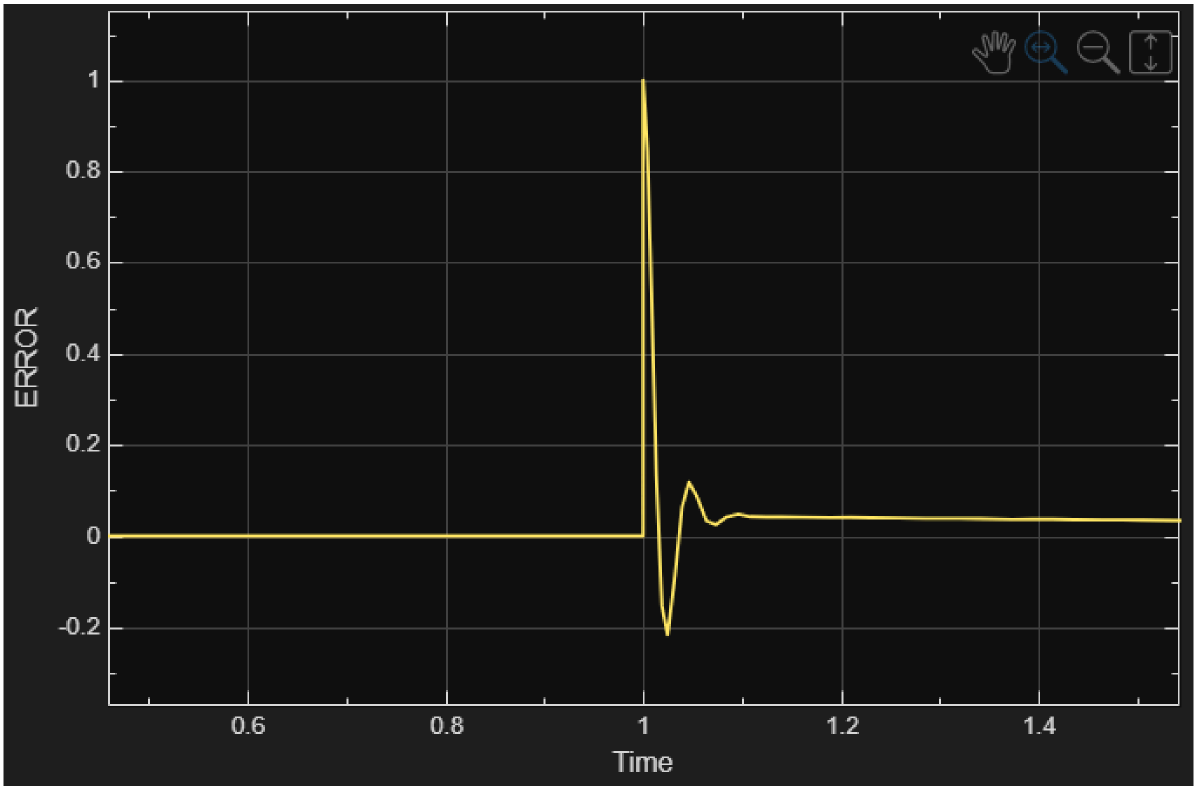


to remove the overshoot (P=0.8 D=2)

Screenshots

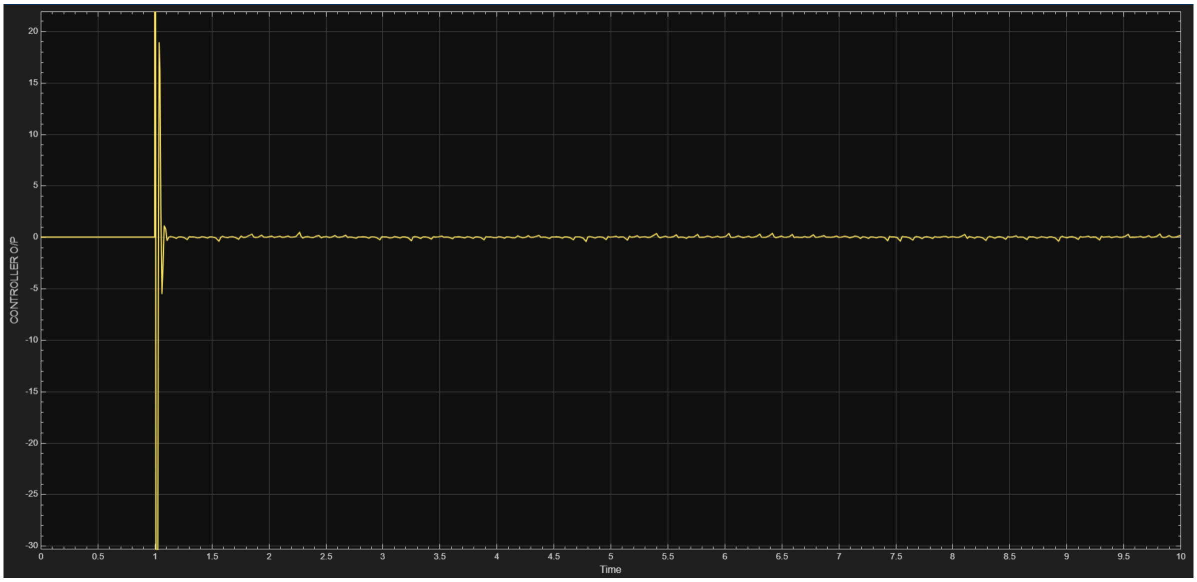
1. ERROR

•



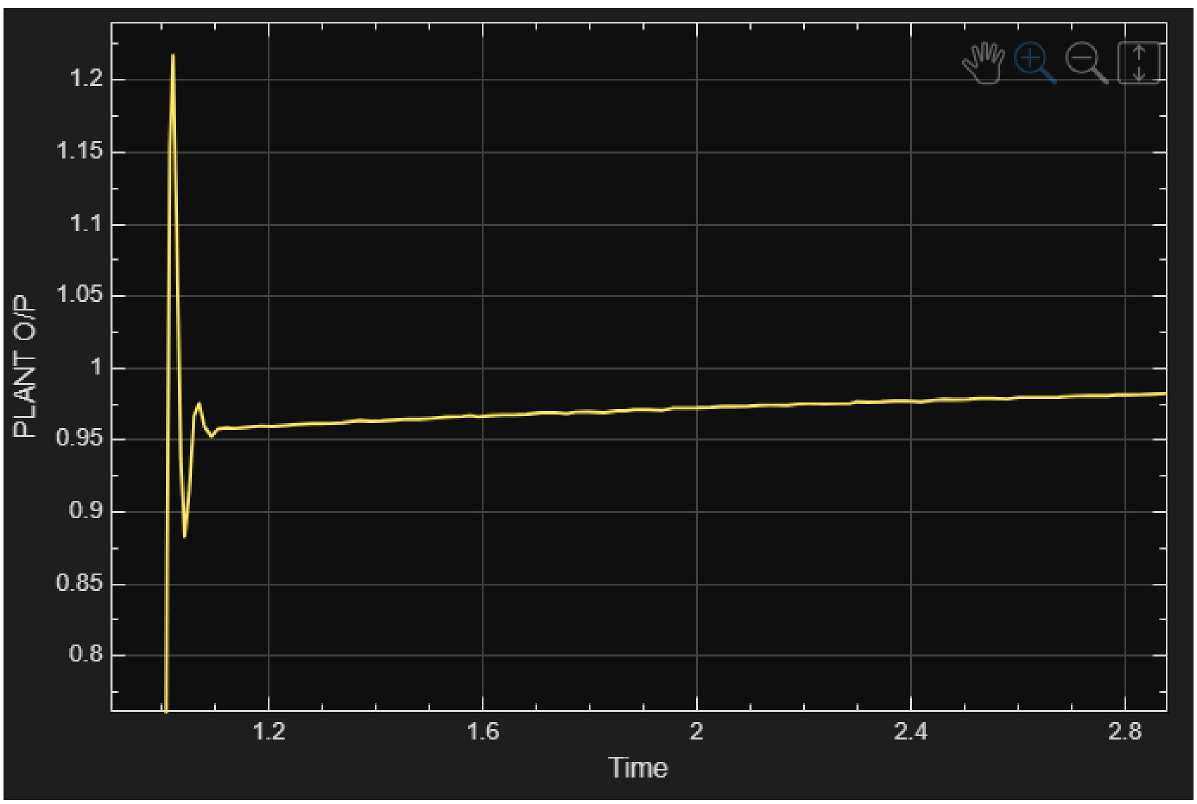
2. CONTROLLER O/P

•



3. PLANT O/P

   -



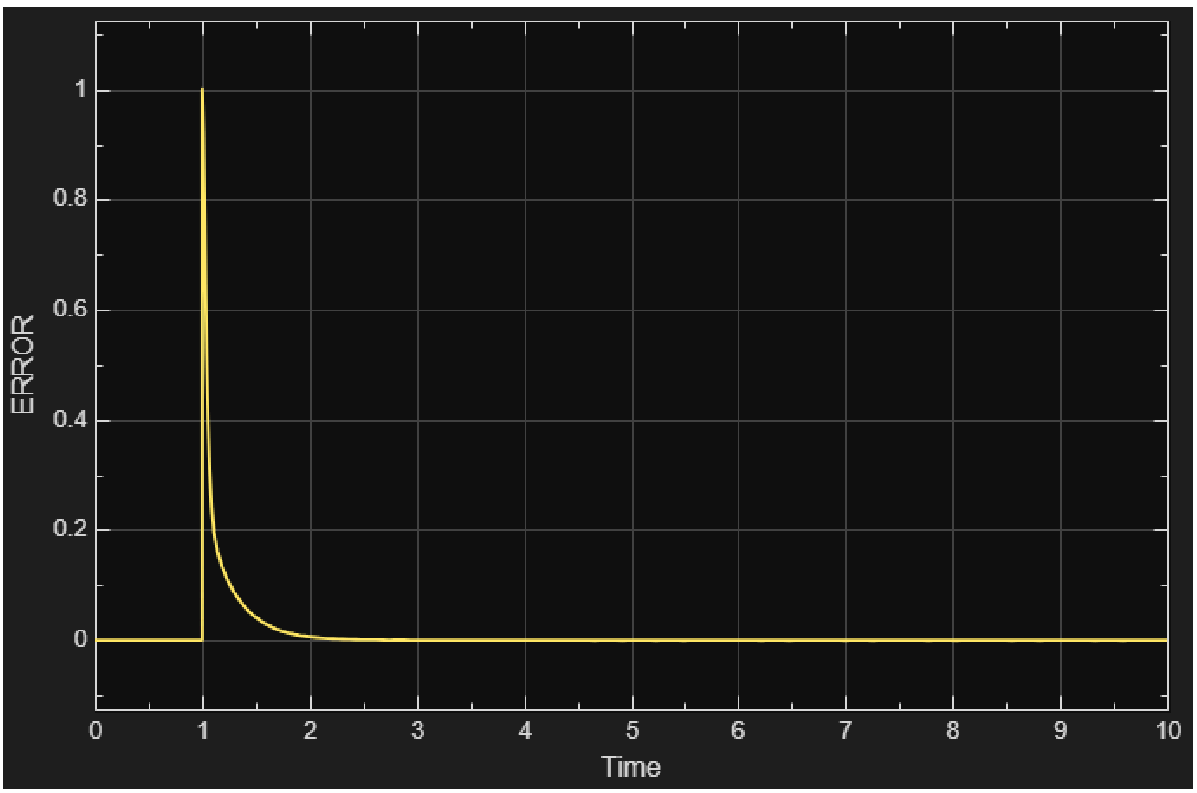
plant output gets more oscillations and overshoots still (and much more than original P=1 case , due to D being too large)

REDUCING D TO CLEAR THE OSCILLATIONS GENERATED AND TO REDUCING THE OVERSHOOT BY PROVIDING DAMPING (P=1 D=0.2)

Screenshots

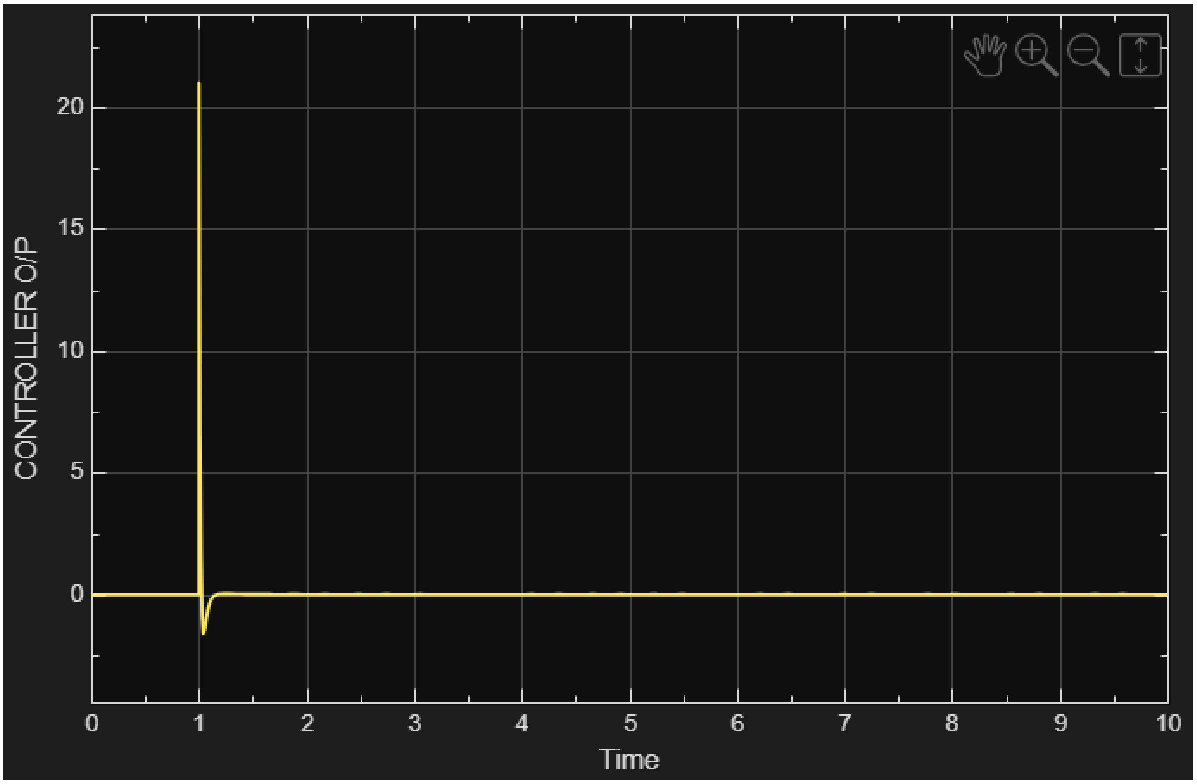
1. ERROR

    •



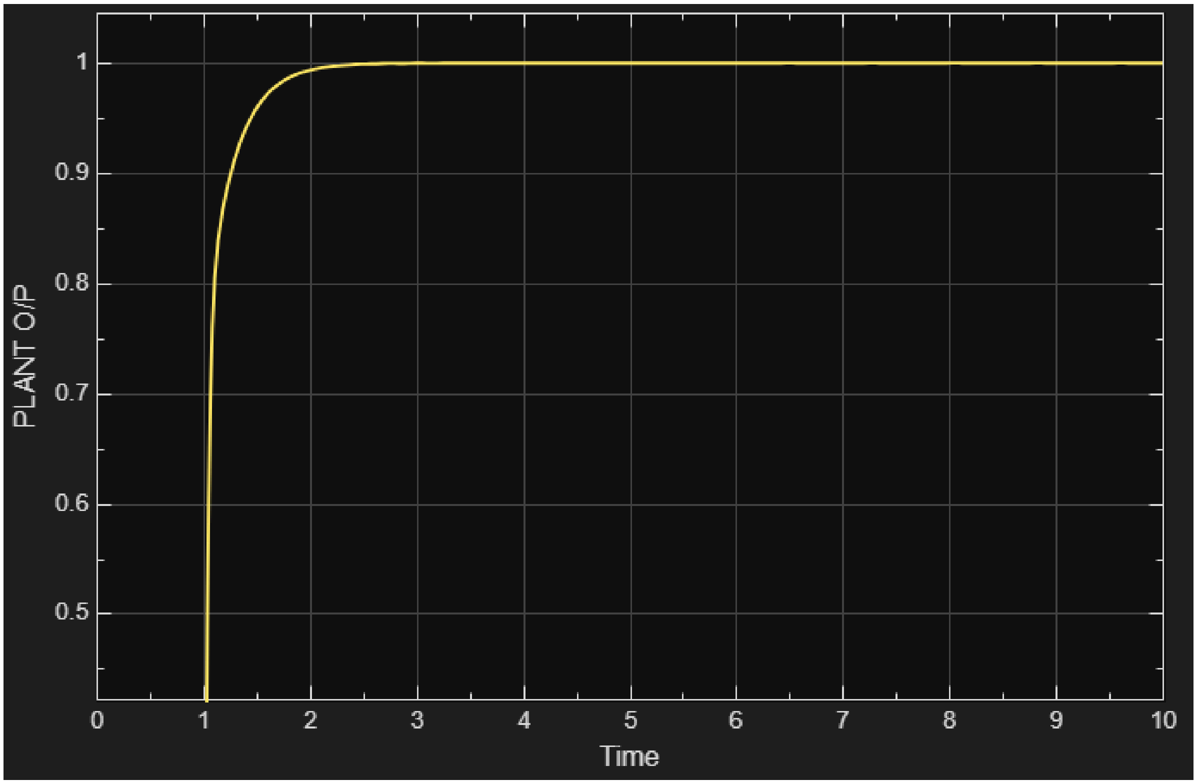
2. CONTROLLER O/P

    •



3. PLANT O/P

-



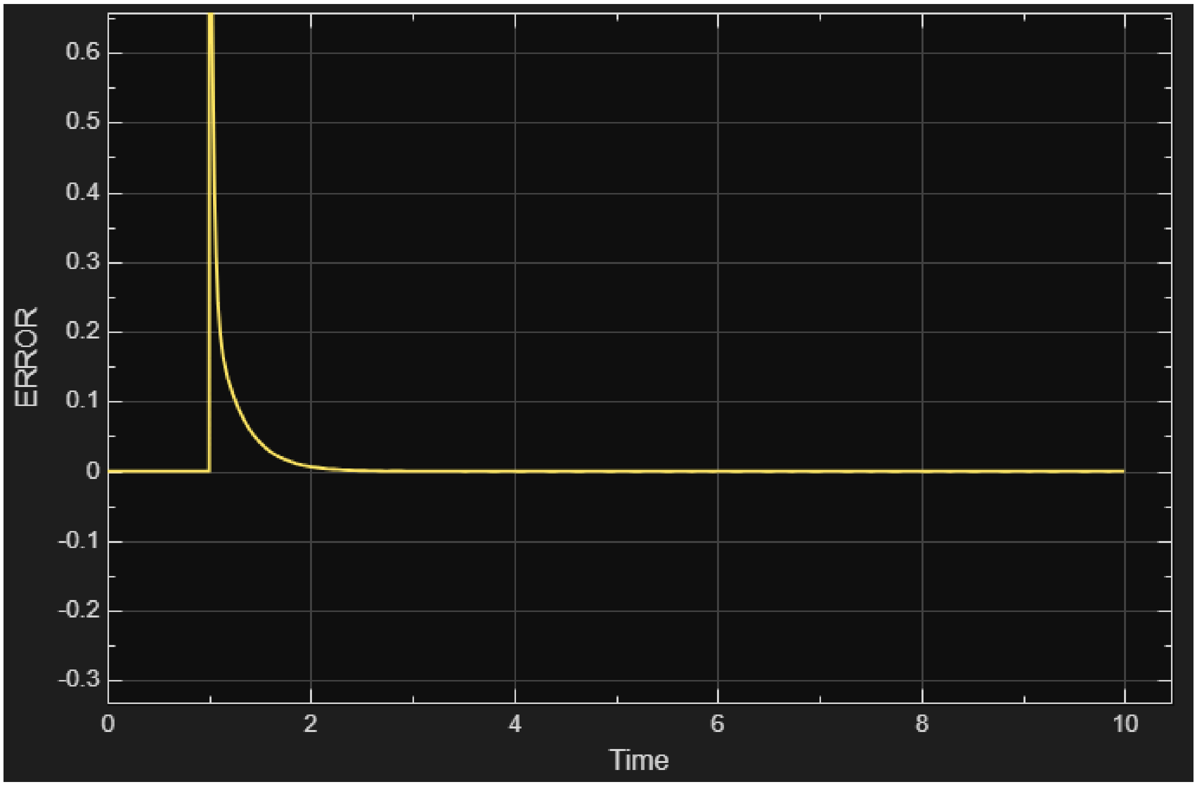
tried it first with D=1 and D=0.7 , calibrated/tuned it to D=0.2 in the end by using trial and error and found a value for D which doesnt give oscillations and also solves the overshoot problem by damping the curve

GIVING I VALUE TO SMOOTHEN OUT THE CURVE

Screenshots

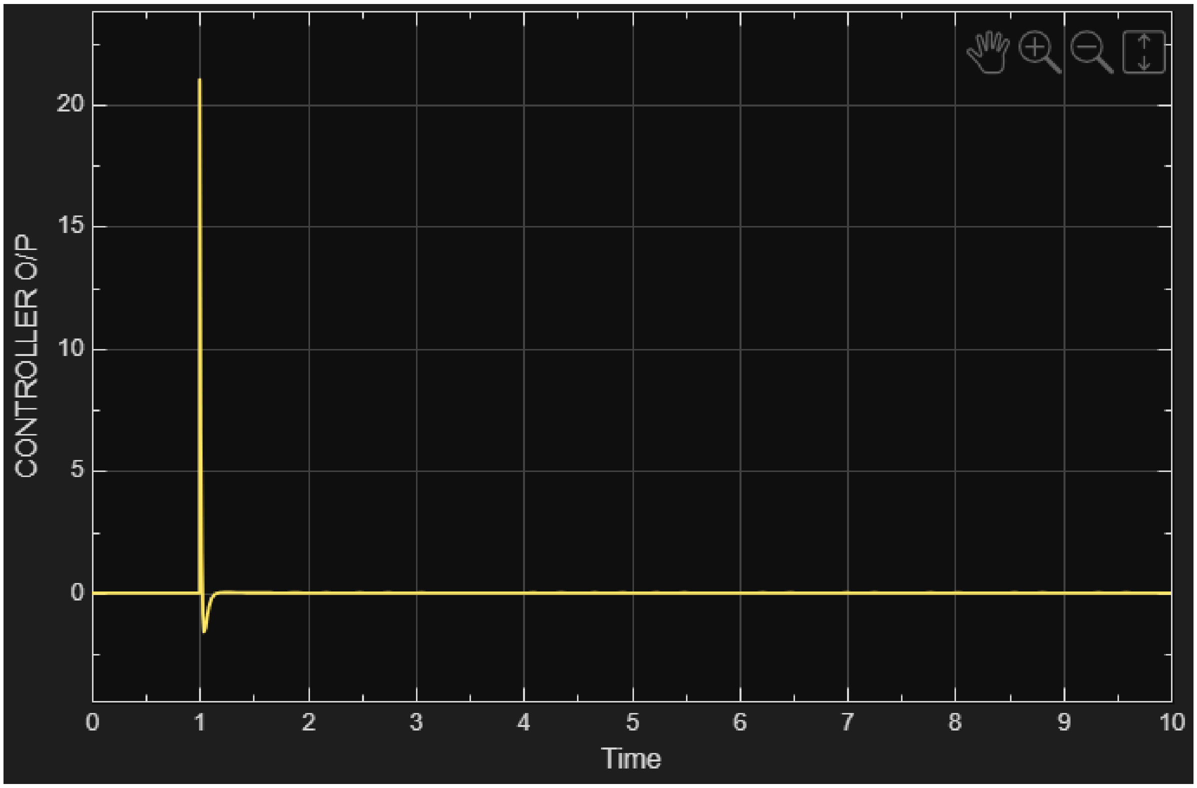
1. ERROR

    -

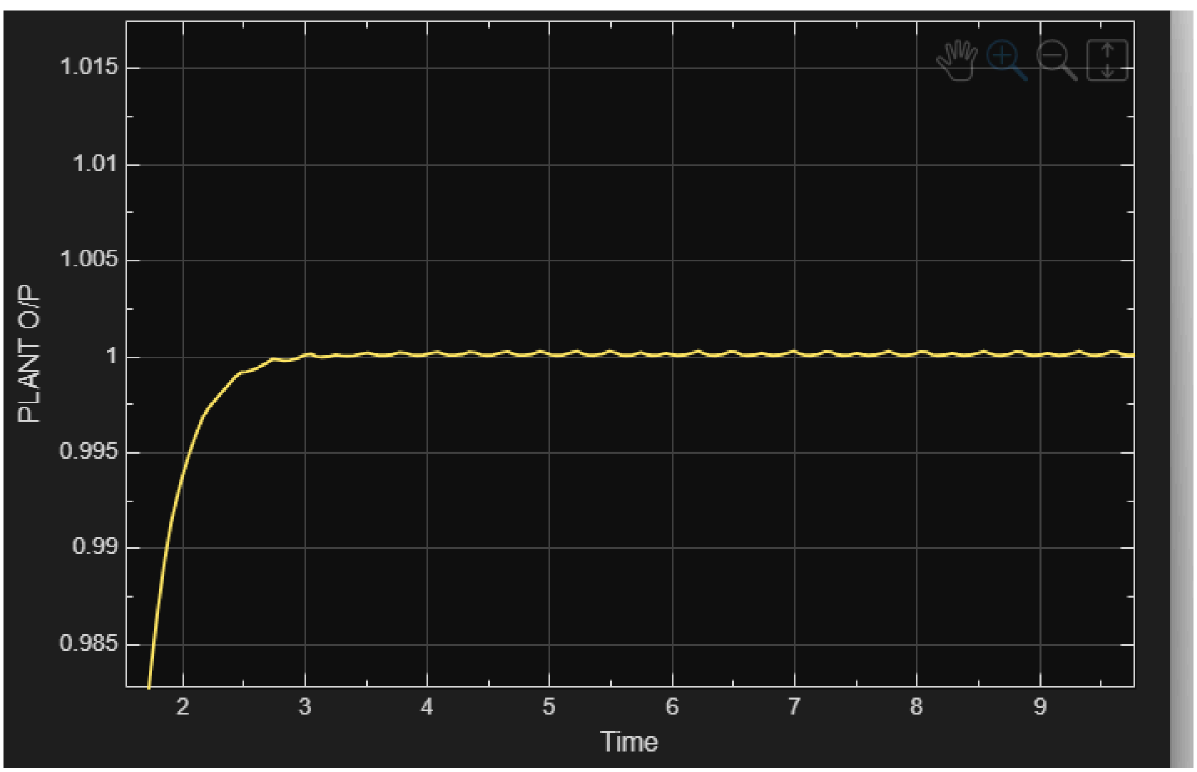


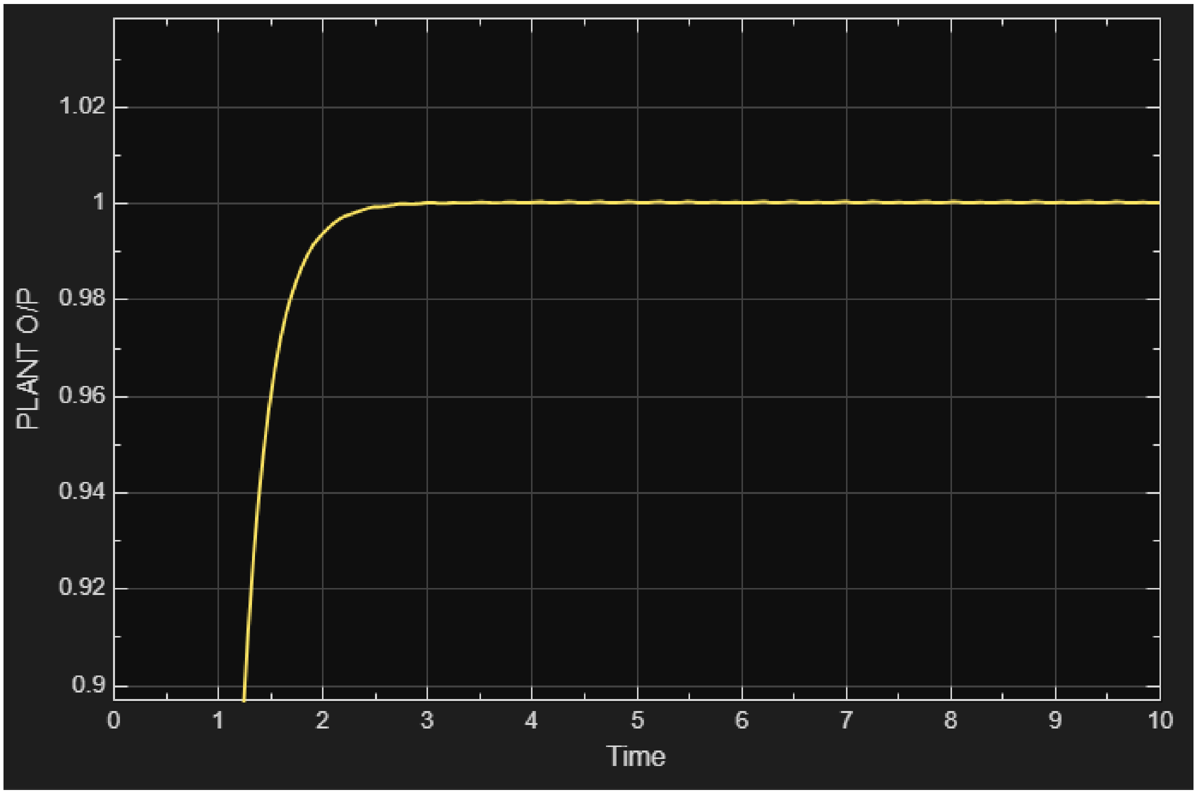
2. CONTROLLER O/P

-

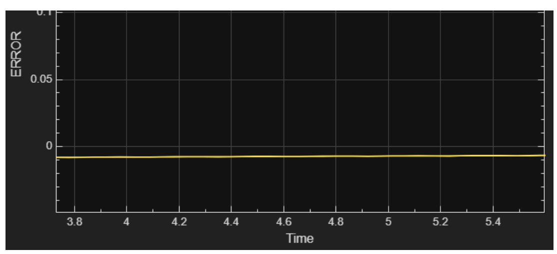

CONTROLLER O/P
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
Time

3. PLANT O/P

- 
PLANT O/P
1.015
1.01
1.005
1
0.995
0.99
0.985
2
3
4
5
6
7
8
9
Time

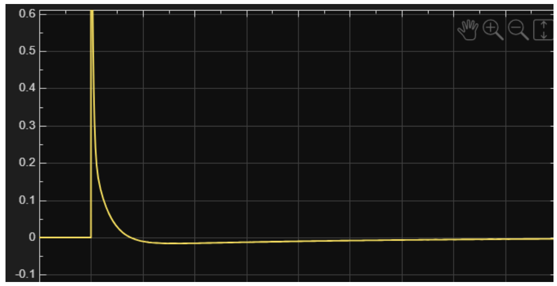


with I=0.1

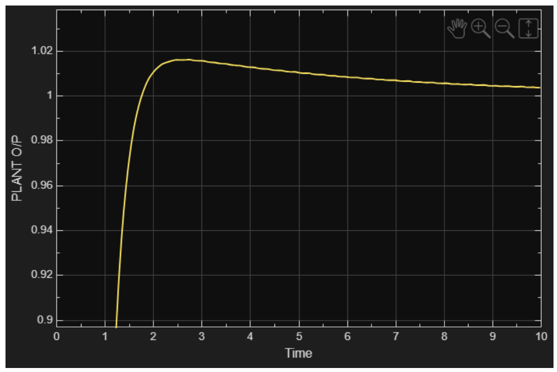


error gets constant in the negative value

with I=0.2



a curve forms which converges towards 0 in the error scope



adds a bit of overshoot , but doesnt exactly reach 1 either for the plant o/p

so for this system i have kept I=0